Notes on David Albert's Lectures on the Foundations of Quantum Mechanics

Notes from the lecture series, expanded and made self-contained

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Abstract

These notes reconstruct and expand the most striking arguments from David Albert's lecture series on the foundations of quantum mechanics (closely following his book *Quantum Mechanics and Experience*). Three threads are covered: (i) the "hardness and colour" box experiments, which establish that quantum randomness is not ignorance and that superposition is a genuinely new mode of being; (ii) the EPR argument and a Bell-type theorem showing that the perfect correlations of entangled pairs cannot be explained by any local pre-assignment of values; (iii) superdeterminism, the escape route from Bell's theorem that denies the theorem's quietest assumption — that the measurement settings are statistically independent of what the particles carry; and (iv) a comparison of the main realist responses — GRW collapse, Bohmian mechanics, and Everettian many-worlds — and what each one costs.

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1 Setting the stage: hardness and colour

Albert's pedagogical device is to strip the spin-$\frac{1}{2}$ degree of freedom of an electron of all its historical baggage and rename it. Electrons are imagined to have two binary properties:

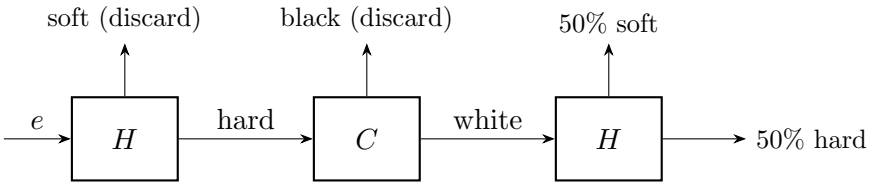
- **Hardness** H: every electron measured for hardness comes out either *hard* or *soft*;
- **Colour** C: every electron measured for colour comes out either *black* or *white*.

Physically, "colour" and "hardness" are spin components along two orthogonal axes (say S_z and S_x), but nothing in the arguments below depends on knowing that. The measurements are performed by *boxes* (think Stern–Gerlach magnets): an electron enters through one aperture and exits through one of two ports labelled by the two possible values. Crucially, these measurements are *repeatable*: a hard electron fed back into a hardness box comes out hard, with certainty, every time. So the boxes deserve to be called measuring devices.

The empirical facts are these: hardness and colour are statistically independent (a white electron is 50/50 hard/soft, and vice versa), and — the interesting part — they are *incompatible*, in the following precise operational sense.

2 Observation 1: measuring colour scrambles hardness

Observation 1. *Take an electron and pass it through a hardness box; select the ones that exit* hard. *Pass these through a colour box; select the ones that exit* white. *Pass these through a second hardness box. One might expect 100%* hard *— these electrons were, after all, just certified hard. Instead one finds 50% hard, 50% soft.*



Measuring colour *scrambles* hardness, and (by symmetry) measuring hardness scrambles colour. You cannot know both at once. This is just the operational face of non-commuting observables, but two features of the scrambling deserve emphasis, because everything later hangs on them.

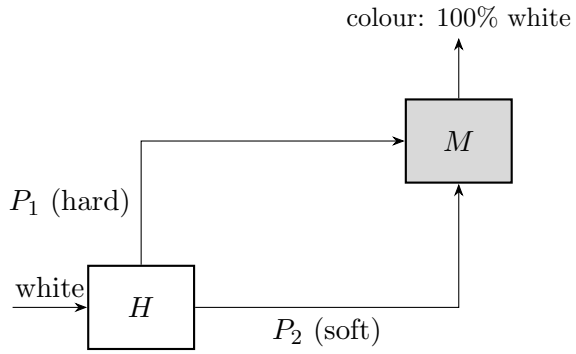
The statistics are immovable. No refinement of the experiment changes the 50/50. You can build the boxes differently, prepare the electrons differently, select sub-ensembles by any measurable criterion whatsoever — *there is nothing one can do to move those statistics one jot.* This is utterly unlike a coin flip, where the 50/50 reflects ignorance of initial conditions and a sufficiently obsessive experimenter could bias the outcome. The quantum 50/50 behaves as if it were a law of nature, not a confession of ignorance. (Later, Bell's theorem will harden this impression into a theorem: the outcomes literally could not have been decided in advance by any local facts about the electron.)

It is not a clumsiness problem. The naive gloss — "the colour measurement physically disturbs the electron and knocks its hardness about" — turns out to be untenable, and Observation 2 is the demonstration.

3 Observation 2: the two-path experiment

3.1 Recombining the paths undoes the scrambling

Now build a hardness box whose two output beams are not measured but instead steered by mirrors into a common region where they are recombined into a single beam (the recombiner does nothing except bring the paths back together; call the hard route P_1 and the soft route P_2). Feed in a *white* electron and measure the *colour* of what comes out.



If the story were "a hardness measurement disturbs colour," then an electron that took either route through this contraption should have its colour scrambled, and the output should be 50/50. It is not:

Observation 2. *Every white electron sent through the recombining device comes out* white*, 100% of the time. The device as a whole does nothing to colour. Hardness "didn't scramble colour," because the potential paths were made to converge in the same place.*

But now do anything that determines *which* path the electron took — put a detector on one arm, or simply block one arm with a wall — and the 50/50 scrambling reappears (with a wall in place, half the electrons are absorbed, and the half that get through are 50/50 black/white). Merely *knowing* the route costs you the certainty of the output colour.

3.2 The formalism in two lines

The two properties correspond to two orthonormal bases of a single two-dimensional Hilbert space, related by

$$|\text{white}\rangle = \tfrac{1}{\sqrt{2}}\big(|\text{hard}\rangle + |\text{soft}\rangle\big), \qquad |\text{black}\rangle = \tfrac{1}{\sqrt{2}}\big(|\text{hard}\rangle - |\text{soft}\rangle\big), \tag{1}$$

(the Born rule then gives all the 50/50s above: e.g. $|\langle\text{hard}|\text{white}\rangle|^2 = \tfrac{1}{2}$). Inside the apparatus, hardness is correlated with route, so a white electron is in the superposition

$$|\psi\rangle \;=\; \tfrac{1}{\sqrt{2}}\big(|P_1\rangle\;|\text{hard}\rangle + |P_2\rangle\;|\text{soft}\rangle\big). \tag{2}$$

If the two routes are recombined *coherently* (equal path lengths, no which-path record anywhere in the universe), the route factors become identical and drop out, leaving $\frac{1}{\sqrt{2}}(|\text{hard}\rangle + |\text{soft}\rangle) = |\text{white}\rangle$: certainty of white is restored by *interference* of the two branches. If instead any degree of freedom ends up correlated with the route — a detector, a stray photon, a wall —

the branches can no longer interfere, and each branch alone gives 50/50 colour. That is what "measurement scrambles the incompatible property" really is: not a mechanical kick, but a destruction of interference.

3.3 The total-of-nothing box, and the collapse of every which-path story

Here is the most philosophically loaded step, which the lectures attribute to a device of Yakir Aharonov's: the *total-of-nothing* (ToN) box. It is a gadget with the following two *measurable* certifications:

1. every measurable property of an electron going in equals the corresponding property going out — hardness, colour, energy, everything — except its position in space (it does come out the other side);
2. the time taken to traverse the box equals the time taken to traverse the same stretch of empty space.

By every operational standard, the box does *nothing at all* to an electron that passes through it. (In the formalism it multiplies the state by -1 — a phase. A global phase has no measurable consequences whatsoever on a single beam, which is exactly why the box can pass both certifications.)

Now insert one ToN box into a *single* arm of the recombining device, say the hard path P_1, and again feed in white electrons:

Observation 3. *With the ToN box on one path, the output is 100%* black.

The one-line calculation: the branch through P_1 acquires the factor -1,

$$\tfrac{1}{\sqrt{2}}\big(|P_1\rangle\,|\text{hard}\rangle+|P_2\rangle\,|\text{soft}\rangle\big) \;\longrightarrow\; \tfrac{1}{\sqrt{2}}\big(-|P_1\rangle\,|\text{hard}\rangle+|P_2\rangle\,|\text{soft}\rangle\big) \xrightarrow{\text{recombine}} \tfrac{1}{\sqrt{2}}\big(-|\text{hard}\rangle+|\text{soft}\rangle\big) = -\,|\text{black}\rangle\,, \tag{3}$$

and $|-1|^2 = 1$: black with certainty. The phase that was unmeasurable on one beam is perfectly measurable as a *relative* phase between two beams.

Now run the classic dilemma. A given electron went in white and came out black, so *something* changed it. Which route did it take?

- **Not P_1 alone.** The ToN box demonstrably changes no property of any electron that traverses it, and the rest of P_1 is empty space. An electron travelling only along P_1 would have come out white.
- **Not P_2 alone.** There is nothing on P_2 but empty space; the *absence* of a box on your own path can hardly turn you from white to black.
- **Not both.** Electrons are never found split; if you look for the electron mid-flight you always find the whole electron on exactly one path, never fragments on each.
- **Not neither.** Block both paths and nothing comes out at all.

Every answer a classical mind can formulate is experimentally refuted. The electron's mode of travel between the boxes was not "P_1," not "P_2," not "both," not "neither." It was something for which classical language simply has no word: a *superposition* of taking P_1 and taking P_2. Albert's point in belabouring this is that superposition is not a fancy way of encoding ignorance about which path was really taken; it is a genuinely novel, experimentally certified mode of being, and any acceptable interpretation of quantum mechanics owes us an account of it.

Punchline. Observation 1 shows quantum statistics are not ignorance about pre-existing values. Observation 2 shows measurement-disturbance stories fail, and that "which value/which path"

questions can lack answers altogether. The measurement problem is then immediate: the linear dynamics that produces these superpositions should, applied to a measuring device (which is just a big collection of electrons and nuclei), produce superpositions of *pointer readings* — yet we always see one reading. Sections 4 and 6 are about the ways out.

4 Lecture 4: EPR and a Bell-type theorem

4.1 The setup: perfectly anticorrelated pairs

There is a preparation protocol P (in real physics: producing two spin-$\frac{1}{2}$ particles in the singlet state) that emits *pairs* of particles, P_1 and P_2, which fly apart to two distant laboratories. Each experimenter may measure any one of three properties A, B, C on their particle, each with possible values ± 1. (In the concrete realization, A, B, C are spin components along three coplanar axes 120° apart.) The defining empirical fact of the protocol:

> If both experimenters measure the *same* property — A on both, B on both, or C on both — the outcomes are *opposite* ($+1$ and -1) 100% of the time.

4.2 The EPR argument: locality forces predetermined values

Suppose **locality**: nothing you do to P_1 can influence P_2 once they are far apart (certainly not faster than light; the measurements can be made at spacelike separation). Measuring A on P_1 then lets you predict with certainty the value of A on distant P_2 — without touching P_2. If the measurement on P_1 cannot have *produced* the opposite value at P_2 (that is what locality forbids), the only remaining explanation of the perfect anticorrelation is that each particle *already carried* definite values of A, B, and C, fixed at the source before the particles separated, with the pair anticorrelated property-by-property. This is the Einstein–Podolsky–Rosen conclusion (1935), in Bohm's spin version: locality ⇒ hidden definite values. Quantum mechanics, which assigns no such values, would then be an incomplete description.

So far this is coherent — even attractive. Bell's contribution (1964; the version below is essentially Mermin's) is that this explanation is not merely unorthodox but *quantitatively wrong.*

4.3 What the pre-assigned values would imply

Suppose each pair leaves the source with definite values. Then each emission is described by a row of a simple table: P_1 carries some triple of ± 1's, and P_2 carries the opposite triple. There are only $2^3 = 8$ possible rows:

P_1			P_2		
A	B	C	A	B	C
$+1$	$+1$	$+1$	-1	-1	-1
$+1$	$+1$	-1	-1	-1	$+1$
$+1$	-1	$+1$	-1	$+1$	-1
$+1$	-1	-1	-1	$+1$	$+1$
-1	$+1$	$+1$	$+1$	-1	-1
-1	$+1$	-1	$+1$	-1	$+1$
-1	-1	$+1$	$+1$	$+1$	-1
-1	-1	-1	$+1$	$+1$	$+1$

Consider the three *mixed* experiments in which the two labs measure *different* properties: (A on P_1, B on P_2), (B on P_1, C on P_2), and (A on P_1, C on P_2). Scan any row of the table

and ask: is at least one of these three pairs opposite? The answer is *yes, in every single row.* The reason is a pigeonhole argument: among the three values A_1, B_1, C_1 carried by P_1, at least two must be equal (three binary values, two slots). Whichever two are equal — say $A_1 = B_1$ — the corresponding mixed pair is opposite, since $B_2 = -B_1 = -A_1$. Hence, on the hypothesis of pre-assigned local values,

$$\Pr\big[\text{at least one of the three mixed pairs is opposite}\big] \;=\; 1 \qquad \text{(every row, every run).} \quad (4)$$

4.4 What quantum mechanics says

Quantum mechanics makes a flat prediction for the mixed experiments. For the singlet state, if the two labs measure spins along axes at angle θ, the probability of *opposite* outcomes is $\cos^2(\theta/2)$. Same axis ($\theta = 0$): probability 1 — the perfect anticorrelation we started from. Different properties, axes 120° apart:

$$\Pr[\text{opposite}] \;=\; \cos^2(60^\circ) \;=\; \tfrac{1}{4}. \quad (5)$$

Then, by the union bound (probabilities of a union of events cannot exceed the sum),

$$\Pr\big[\text{at least one of the three mixed pairs is opposite}\big] \;\le\; \tfrac{1}{4} + \tfrac{1}{4} + \tfrac{1}{4} \;=\; \tfrac{3}{4}. \quad (6)$$

(One cannot measure all three mixed pairs on a single pair of particles, of course; (6) is a statement about the counterfactual values the local story insists exist, and it is checkable pair-by-pair by choosing the settings at random.)

Equations (4) and (6) contradict each other: $1 \not\le \frac{3}{4}$. And experiment — Freedman–Clauser, Aspect, and the 2015 loophole-free tests (2022 Nobel Prize) — sides with quantum mechanics.

4.5 What must go

The pre-assigned values were not an optional flourish; they were *forced* by locality via the EPR argument. So the conclusion is stark:

> **Locality is false.** The correlation between the two outcomes does not propagate; it does not depend on anything in the space between the labs, and it does not weaken with distance. And there was *no fact of the matter* about the outcomes prior to the experiment — the values are not carried from the source. The world is, in this specific sense, non-local.

(This is the verdict *given* two background assumptions that the argument never states out loud: that each measurement has a single outcome, and that the experimenters' choice of what to measure is statistically independent of which row of the table the source picked. The first is denied by Everett, below; the second is denied by superdeterminism, §5.)

Note what is *not* implied: no signalling. The local statistics in either lab are 50/50 regardless of what the other lab does, so the non-locality cannot be used to communicate. But as an account of what nature is doing, locality in Einstein's sense is dead.

The many-worlds escape hatch. There is one loophole that Albert flags repeatedly: the entire argument presupposes that each measurement has a *unique, determinate outcome* — that "the probability the outcomes were opposite" refers to something. In the Everett interpretation this is denied: both outcomes occur, in different branches, and there is no unique fact of the matter about how the experiments turned out. The correlations, as single-world facts, simply don't exist; what exists is a branching structure in which they appear. With that premise removed, Bell's theorem cannot get started, and many-worlds remains a fully *local* theory. This is one of its principal selling points — at a price described below. There is one further escape, which keeps single outcomes, keeps determinism, keeps locality, and pays for it in a different currency altogether. It gets a section of its own.

5 Superdeterminism: the assumption nobody wrote down

This section goes beyond the lectures. Bell himself named the escape route discussed here, in a 1985 BBC radio interview with Paul Davies:

> "There is a way to escape the inference of superluminal speeds and spooky action at a distance. But it involves absolute determinism in the universe, the complete absence of free will. Suppose the world is super-deterministic, with not just inanimate nature running on behind-the-scenes clockwork, but with our behavior, including our belief that we are free to choose to do one experiment rather than another, absolutely predetermined, including the 'decision' by the experimenter to carry out one set of measurements rather than another, the difficulty disappears."

Bell mentioned it in order to dismiss it, and for decades that was the standard attitude. It has since had a small but serious revival, and it is worth seeing exactly which step of §4 it attacks.

5.1 Where the assumption hides

Go back to the table of eight rows. The contradiction came from setting two things side by side: (4), a fact about *every row*, and (6), which adds up three probabilities measured in *three different sets of runs* — the runs in which the labs happened to measure (A, B), those in which they measured (B, C), and those in which they measured (A, C). To compare the two we silently assumed that each of these sets of runs is a *fair sample* of the rows the source emits: that the source does not emit a different mix of rows on days when the labs are going to measure (A, B) than on days when they are going to measure (B, C). That is what the parenthetical "choosing the settings at random" was doing.

Written out in general, let λ stand for whatever the source gives the pair (a row of the table, or anything richer), distributed with density ρ, and let a, b be the settings chosen in the two labs. Bell's theorem needs two premises:

$$\text{Locality:} \quad p(x, y \mid a, b, \lambda) = p(x \mid a, \lambda)\, p(y \mid b, \lambda), \tag{7}$$

$$\text{Statistical independence:} \quad \rho(\lambda \mid a, b) = \rho(\lambda). \tag{8}$$

The first says each outcome depends only on the local setting and on what the particle brought with it. The second — also called *measurement independence*, or the *free-choice* assumption — says what the particles brought with them is uncorrelated with what they will be asked. A **superdeterministic** theory keeps (7) and denies (8).

5.2 How cheap the escape is

Once (8) is dropped, reproducing the quantum statistics with purely local, pre-assigned values is embarrassingly easy. Suppose the source "knows" the pair of settings $(x, y) \in \{A, B, C\}^2$ that will be used on this run, and picks a row as follows:

1. give P_1 a uniformly random value ± 1 of property x;
2. if $y \neq x$: with probability $\frac{1}{4}$ give P_1 the *same* value of y as of x, otherwise the opposite value; fill in the third property arbitrarily;
3. give P_2 the opposite triple, as always.

Check it: same-setting runs are always opposite (every row does that). For a mixed run, the outcomes are x_1 and $y_2 = -y_1$, which are opposite exactly when $y_1 = x_1$ — probability $\frac{1}{4}$, precisely the quantum $\cos^2 60°$. Each lab sees 50/50, whatever the other lab does, so nothing can be signalled. And every particle is a perfectly classical object carrying a complete row of definite values, read off locally. The pigeonhole argument is still true of every row; it just

no longer matters, because the rows emitted on (A, B)-days are not the rows emitted on (B, C)-days.

All the difficulty has been moved into the word "knows". The settings are chosen *after* the pair has left the source, in labs that may be light-years apart, by whatever process the experimenters like. For the source's row to be correlated with them, the correlation must be laid down in the common past of the source and both setting-choosers — and it must hold *whatever that process is*.

5.3 Why "super"? Determinism is not enough

A merely deterministic world (Laplace's clockwork, or Bohmian mechanics) does not violate (8). In such a world the settings and λ are both fixed by the initial state of the universe, but in the ordinary way that two well-shuffled decks are both fixed by physics yet are uncorrelated with each other. Bell made the point in 1977 with a pseudo-random number generator: its output depends so sensitively on so many incidental details (say, the parity of the millionth digit of some input) that, although fully determined, it is as good as free for the purpose of choosing a setting. Superdeterminism is the much stronger claim that the initial conditions and laws are such that settings and λ are correlated in *just* the right way, for every setting-choosing mechanism anyone will ever use.

For the same reason modern proponents, notably Hossenfelder and Palmer, argue that the phrase "free will" is a red herring: no human freedom is involved when the setting is chosen by a noisy diode. The substantive claim is simply the violation of (8).

A backward look at Observation 1. Recall the claim that quantum 50/50s are "immovable": no sub-ensemble selected by any measurable criterion shifts them, so they cannot be ignorance. That inference, too, assumed our selections are independent of the hidden state of the electron. A superdeterminist says the 50/50 *is* ignorance after all — we just cannot select against it, because our selecting is part of the same correlated history.

5.4 Experiments that push the conspiracy back in time

One cannot rule out a correlation laid down at the Big Bang, since every past light-cone overlaps there. But one can rule out any mechanism that establishes it *recently*, by choosing the settings with processes causally disconnected from the source for as long as possible:

- **Human choices** (BIG Bell Test, 2018): roughly 100,000 volunteers playing an online game supplied the setting bits to twelve laboratories. Local realism was violated anyway.
- **Starlight** (Handsteiner et al. 2017): settings determined by the colour of photons from Milky Way stars, emitted about 600 years earlier.
- **Quasars** (Rauch et al. 2018): settings determined by photons from two high-redshift quasars, emitted about 7.8 and 12.2 billion years ago, with a Bell violation of 9.3 standard deviations. Any correlation between settings and λ must now be at least 7.8 billion years old.

These do not refute superdeterminism; they fence it in. A related result shows the fence cannot be made tight by weakening alone: Hall (2010) built a local, deterministic, non-signalling model reproducing *all* singlet correlations while giving up only a fraction (about 14%, on his measure) of measurement independence. The correlation need not be total, merely present.

5.5 The case against

- **Conspiracy / fine-tuning.** The correlation must be exactly strong enough to mimic quantum statistics, exactly weak enough never to permit signalling, and indifferent to

whether the setting came from a diode, a human or a quasar. Such precision, with no mechanism in sight, looks like an explanation that could accommodate *anything*. Sen and Valentini (2020) argue that superdeterministic models need finely tuned initial conditions, and that generic departures from them would permit signalling; Wood and Spekkens (2015) show that *every* causal account of Bell correlations requires fine-tuning of some kind, so this cost is not unique to superdeterminism, but here it is especially stark.

- **The end of experimental science?** Shimony, Horne and Clauser (1976) observed that every controlled experiment — a drug trial, say — assumes that the randomising device assigning patients to treatment is uncorrelated with their prognosis. If nature may correlate settings with hidden variables whenever it likes, controlled inference collapses everywhere. Proponents reply that the violation of (8) is confined to the microscopic degrees of freedom involved in quantum measurement, and that no such correlation is posited for pills and patients; critics ask what principle enforces the confinement.

- **No theory yet.** There is no complete superdeterministic theory that reproduces quantum mechanics. What exists is a programme: 't Hooft's cellular-automaton interpretation; Palmer's invariant set theory, in which the counterfactual settings needed for the pigeonhole argument lie off a fractal set of allowed states, so the eight-row table is not even well-defined; and toy models such as Donadi and Hossenfelder's (2022) local, deterministic account of collapse.

5.6 Could it be tested?

Unlike the other interpretations, superdeterminism predicts that quantum randomness is not fundamental, which suggests where to look. Hossenfelder has proposed that if outcomes are fixed by hidden variables that include the state of the measuring device, then repeating a measurement on a small, cold, well-isolated system, with the same settings and in rapid succession (before the hidden variables have time to change), should show time correlations between outcomes that the Born rule forbids. No such deviation has been reported, but the test has not yet been done in the regime the proposal requires. On the theoretical side, recent work (e.g. Waegell, 2026) tries to give a precise criterion for when a theory counts as *non*-superdeterministic, so that the charge of "conspiracy" can be made quantitative rather than rhetorical.

A cousin: retrocausality. (8) can also fail for a causal, rather than conspiratorial, reason: if the future settings influence the state of the pair *backwards in time* along their worldlines. Retrocausal models (Price; Wharton and Argaman, 2020) violate (8) while remaining local in spacetime, trading "conspiracy in the initial conditions" for "causation that runs both ways".

Punchline. Bell's theorem is a theorem: locality plus statistical independence plus single outcomes contradicts experiment. Albert's lectures give up locality (GRW, Bohm) or single outcomes (Everett). Superdeterminism gives up statistical independence and keeps everything else — locality, determinism, single outcomes, even Einstein's hope of values existing before measurement. The price is paid not in exotic ontology but in initial conditions: the universe must be arranged so that what we choose to ask the particles is correlated with what they already carry, however we make the choice. Whether that is a deep new principle or a conspiracy theory is exactly what is in dispute.

6 The interpretations: what each one buys and what it costs

All parties agree on the phenomena and (in the non-relativistic domain) on the statistics. They disagree about what is really going on — in particular about what happens in a "measurement," given the superposition arguments of Observation 2. The three realist programmes discussed in the lectures:

	Pros	**Cons**
GRW (Schrödinger evolution + stochastic jumps)	Solves the measurement problem with explicit, precise dynamics; in principle empirically distinguishable from standard QM.	There truly is *no fact of the matter* about uncertain quantities until a jump occurs; until a big system gets involved, superpositions are ontologically real. Work needed for a relativistic version. Non-local.
Bohm (pilot wave)	Ontologically conservative: there are only waves and particles; particles always have definite positions; no collapse, no exotic metaphysics of outcomes.	The wave lives in $3N$-dimensional configuration space (N = number of particles in the universe), so the "wave" is nothing like a field in 3-space. Work needed for a relativistic version. *Wildly* non-local.
Everett (many worlds)	Local. Just the Schrödinger equation, nothing else — no collapse, no extra variables. Generalises to relativistic QFT trivially.	Unclear what probability means: in what sense can both worlds exist, with certainty, and yet have different probabilities associated with them?
Superdeterminism (not in the lectures; §5)	Local, deterministic, single outcomes, definite values before measurement. Quantum randomness becomes ignorance, so it is in principle testable.	Settings must be correlated with hidden variables however they are chosen: looks conspiratorial and threatens controlled inference generally. No complete theory reproducing quantum mechanics yet exists.
Bohr (Copenhagen)	—	Impossible to tell what the view actually says is going on. The stories above are a proof by construction that he was wrong that no such story could be told.

6.1 GRW: make the collapse real

Ghirardi, Rimini and Weber (1986) modify the Schrödinger equation itself: each elementary particle has a tiny probability per unit time (about once per 10^{16} s) of undergoing a spontaneous, random localization ("jump") of its wavefunction. For one electron in an interferometer this essentially never happens, so all the interference phenomena of Observation 2 survive. But a macroscopic pointer contains $\sim 10^{23}$ entangled particles, and one jump anywhere collapses the whole assembly — so superpositions of pointer positions die within microseconds. Measurements have unique outcomes because the dynamics says so.

The costs, per the lecture: between jumps the superposition is *ontologically real* — there is genuinely no fact of the matter about where the electron is, or which path it takes, until either a jump happens or a big system gets entangled with it. The theory is explicitly non-local (a jump here instantaneously reshapes the wavefunction everywhere — this is how it accommodates Bell). And making it relativistic is a serious open research programme, since "instantaneous collapse" picks out a preferred frame. A compensation worth remembering: GRW differs *in principle* from quantum mechanics (interference of sufficiently large systems is predicted to fail), so it is a physical hypothesis, not just a reinterpretation.

6.2 Bohm: particles all the way down

In Bohmian mechanics (de Broglie 1927, Bohm 1952) there are two things: the wavefunction, evolving by the Schrödinger equation always (no collapse, ever), and actual point particles with definite positions at all times, whose velocities are determined by the wavefunction (the "pilot wave"). The electron in Observation 2 really does take exactly one path — but the empty branch of the wavefunction still travels the other path, and recombines to steer the particle, which is how the ToN box on the path *not taken* can flip the outcome. Measurements have unique outcomes because particle configurations are unique; the 50/50 statistics arise from ignorance of the initial particle positions (distributed as $|\psi|^2$). This is the ontologically conservative option: only waves and particles, nothing modal or exotic.

The costs: the "wave" is a function on $3N$-dimensional configuration space — one axis for every coordinate of every particle in the universe — so its claim to be a physical field in space is strained; either you accept a $3N$-dimensional arena as fundamental or you owe a story about how 3-space emerges. The dynamics is *wildly* non-local: the velocity of each particle depends instantaneously on the positions of all the others, however distant (again, this is how Bell is accommodated, and again it requires a preferred foliation of spacetime, making a relativistic version hard).

6.3 Everett: take the Schrödinger equation literally

Everett (1957) keeps the wavefunction and the linear dynamics and adds *nothing*. When a measuring device interacts with the superposed electron, the device — and the experimenter, and the environment — simply join the superposition. Decoherence makes the branches dynamically autonomous, and each branch contains an experimenter who saw a definite outcome. Nothing ever collapses; "both things happen."

The pros are exactly the other theories' cons: the theory is *local* (it evades Bell by denying unique outcomes, as in §4); its dynamics is just the Schrödinger equation, with no ad hoc additions; and since it adds no structure that would conflict with Lorentz invariance, it generalises to relativistic quantum field theory trivially.

The con is the *probability problem*, and Albert regards it as serious. Quantum mechanics is confirmed by its statistical predictions — the Born-rule frequencies. But in Everett, when you measure the hardness of a white electron, *both* outcomes occur with certainty. In what sense, then, was the probability of "hard" equal to $\frac{1}{2}$? What are the relative frequencies evidence *of*, if every run yields every outcome? Unless amplitudes can be given a genuine probabilistic (or decision-theoretic) role, the very evidence that supports quantum mechanics seems to be unstatable within the interpretation. Proposed answers exist (self-locating uncertainty, Deutsch–Wallace decision theory); whether any succeeds is still contested.

6.4 Bohr, by way of epilogue

The Copenhagen position, on Albert's telling, combined two claims: that the quantum formalism needs no realist story beneath it, and that no coherent such story is *possible* — talk

of what the electron is doing between measurements was to be renounced as meaningless. The verdict of the lectures is blunt: whatever one thinks of GRW, Bohm, and Everett, each is a fully coherent, empirically adequate account of what is going on — including during measurements. They are a *proof by construction* that Bohr was wrong. The live question in the foundations of quantum mechanics is not *whether* such a story can be told, but *which* of the tellable stories is true — and each candidate exacts a remarkable price: real chancy collapses (GRW), a $3N$-dimensional pilot wave with radical non-locality (Bohm), or a branching multiverse in which probability itself needs reconstruction (Everett).

Further reading

- D. Z. Albert, *Quantum Mechanics and Experience*, Harvard UP (1992) — the book the lectures track; chapters 1–2 are the hardness/colour experiments, chapter 3 the EPR/Bell argument.
- J. S. Bell, *Speakable and Unspeakable in Quantum Mechanics*, Cambridge UP (2004) — especially "Bertlmann's socks and the nature of reality."
- N. D. Mermin, "Is the moon there when nobody looks?", *Physics Today* **38**(4), 38 (1985) — the version of Bell's theorem used in §4.
- T. Maudlin, *Quantum Non-Locality and Relativity*, 3rd ed., Wiley (2011) — what Bell does and does not imply about relativity.
- D. Wallace, *The Emergent Multiverse*, Oxford UP (2012) — the state of the art on Everettian probability.
- P. C. W. Davies and J. R. Brown (eds.), *The Ghost in the Atom*, Cambridge UP (1986) — the Bell interview in which "super-deterministic" appears.
- J. S. Bell, "Free variables and local causality" (1977), reprinted in *Speakable and Unspeakable*; with A. Shimony, M. A. Horne and J. F. Clauser's "Comment on the theory of local beables" (1976), the original exchange on the free-choice assumption.
- S. Hossenfelder and T. N. Palmer, "Rethinking superdeterminism", *Frontiers in Physics* **8**, 139 (2020), arXiv:1912.06462 — the modern case for, including proposed tests.
- M. J. W. Hall, "Local deterministic model of singlet state correlations based on relaxing measurement independence", *Phys. Rev. Lett.* **105**, 250404 (2010).
- The BIG Bell Test Collaboration, "Challenging local realism with human choices", *Nature* **557**, 212 (2018); D. Rauch et al., "Cosmic Bell test using random measurement settings from high-redshift quasars", *Phys. Rev. Lett.* **121**, 080403 (2018).
- I. Sen and A. Valentini, "Superdeterministic hidden-variables models I: non-equilibrium and signalling", *Proc. R. Soc. A* **476**, 20200212 (2020) — the fine-tuning case against.
- K. B. Wharton and N. Argaman, "Colloquium: Bell's theorem and locally mediated reformulations of quantum mechanics", *Rev. Mod. Phys.* **92**, 021002 (2020) — retrocausal and related escapes.
- G. 't Hooft, *The Cellular Automaton Interpretation of Quantum Mechanics*, Springer (2016).